

Q: 2(i)

A: 2(i)

- ① It should be a general statement.
- ② It should be a tentative idea.
- ③ It should agree with available observations.
- ④ It should be testible and potentially falsifiable.

Q: 2(ii)

A: 2(ii)

* Living characteristics of viruses are:-

- ① They occur in different varieties.
- ② They have their own genetic material in the form of either RNA or DNA.

* Non-living characteristics of viruses are:-

- ① They do not respire.
- ② They can be crystallized or stored in bottle.
- ③ They lack cellular structure and enzyme system.

Viruses are at the borderline of living or non-living characteristic. So they aren't included in any domain.

or Kingdom under modern classification.

Q: 2(iii)(or)

A: 2(iii)(or)

Autotrophs:-

Organisms that are capable of producing their own food are called Autotrophs (~~from~~ photosynthetic mode of nutrition)

Example :- Kingdom Plantae, Protista. ~~For~~ These are producers.

* Heterotrophs:-

Organisms ~~that~~ are which eat other things as their food are called heterotrophs (ingestive mode of nutrition) Example :- Animalia (Kingdom). These are consumers.

* Saprotrophs:-

Organisms which are depend on dead-decaying matter are called saprotrophs (absorptive mode of nutrition) Example :- Fungi, Bacteria. These are decomposers.

~~gas~~ goose

Q.2(iv)

A:2(iv)

The three types of muscle cells are:-
① Smooth muscles:-

in the throat, and in the wall of the gut, where it produces the peristaltic contraction that move food along.

② Striated muscles:-

in the muscles of arms and legs, which move body parts.

③ Cardiac muscles:-

in the heart.

Muscle cells have an elongated shape and are filled with actin and its associated contractile proteins, which is what allows them to contract and relax.

Q.2(v)(or)

Metaphase I :-

the bivalents move to the equatorial plate of the spindle. 1 member of each homologous pair is attached to a spindle fibre.

coming from 1 pole, and the other member to a fibre coming from the opposite pole.

Anaphase I :-

Two homologous chromosomes of each pair separate from each other and move to opposite pole. Each chromosome consists of 2 chromatids and each pole receives a haploid set of chromosomes.

If the parent cell has $2n = 12$, then the haploid number is 6 so each pole receives 6 chromosomes at the end of anaphase I.

Q:2(vi)

E: Interphase

D: Prophase

A: Metaphase

B: Anaphase

C: Telophase

———— Q:2(vii)(or) ————

———— A:2(vii)(or) ————

Muscle tissue:-

contracts and relaxes to produce movement of the body and of materials inside it.

Epithelial tissues:-

Covers and lines the body surfaces and the surfaces of organs, giving protection and forming a barrier.

Nerve tissue:-

Conducts nerve impulses and so carries out coordination and control in the body.

Connective tissue:-

Joins, supports and binds the other tissues and organs together.

———— Q:2(viii) ————

———— A:2(viii) ————

Dermal tissue:-

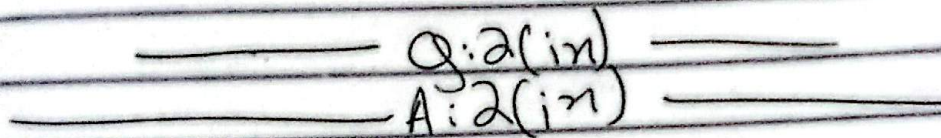
Covers and protect the plant, forming its outer covering.

Ground tissue :

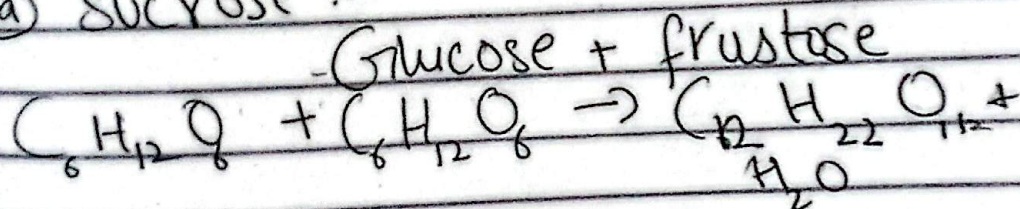
Serves as the site of photosynthesis, and also stores food and provides support.

Vascular tissue :

~~Through the~~ ~~the~~ transports material through the plants. Xylem carries water and dissolved minerals, and phloem carries prepared food.



(a) Sucrose :-



(b) Maltose :-

Glucose + Glucose

(c) Lactose :-

galactose + glucose

Q: 2(n) or —
A: 2(n) or —

Translation takes place in the cytoplasm. The mRNA synthesized in the nucleus moves out into the cytoplasm, where it binds to a ribosome.

Role of tRNA (transfer ribonucleic acid)
It transfers amino acids to the ribosome according to the sequence of codes on the mRNA. Different types of tRNA carry different amino acids, so the right amino acids arrive for each code.

The ribosome reads (translates) the code carried by the mRNA and links the amino acids together, ~~putting~~ producing a polypeptide chain with a specific sequence of amino acids ~~that~~ that is a protein.

Q: 2(xii)

A: 2(xi)

A co-factor is the non protein component that many enzymes need in order to function properly. There are 3 types:-

1) Activator:-

A detachable inorganic ion. Example:- Zn^{++} , Fe^{++} etc, chloride ions.

2) Prosthetic group:-

An organic molecule tightly or permanently attached to the enzyme. For Example FAD, haem group.

3) Coenzyme:-

A detachable organic molecule. For Example:- NAD, vitamin A and C.

Q.3(i)(08)

A:3(i)(08)

(a) A hypothesis should be a general statement.

It should be a tentative idea.
It should agree with an available observation, it should be testable and potentially falsifiable.

(b) A deduction is a logical consequence of a hypothesis. The hypothesis is taken as true and the observer works out what must follow from it, using "if ... then logic".

Worked example:-

Hypothesis :- "malaria is caused by plasmodium". Then the deduction will be:-

"If malaria is caused by Plasmodium then all malarial patients should have Plasmodium in their blood?"

(c) Experimentation is called the most important step bcz it is the only step that actually settles whether the hypothesis

true or false. The deductions drawn from the hypothesis are subjected to rigorous testing and it is through experimentation that the biologist learns which hypothesis is correct. Without it a hypothesis stays no longer more than a plausible guess.

Q: 3(ii)

A: 3(ii)

Meiosis II is similar to mitosis and has four stages:-

① Prophase II :-

the chromatin coils up so that chromosomes become visible again, the spindle apparatus is organized and the nuclear envelope disintegrates.

② Metaphase II :-

the chromosomes line up separately (not in homologous pairs) at the equator of the spindle, and each chromosome is attached by a spindle fibre.

from both poles.

Anaphase II :-

The centromere divides and the sister chromatids are pulled apart to opposite poles.

Telophase II :-

The chromatids reach the pole and uncoil into the chromatin fibres and a nuclear envelope forms each around each set. Cytokinesis II then divides the cytoplasm.

At the end of meiosis II there are 4 daughter cells, and they are haploid (n), each carrying half the chromosome number of the parent cell. Because of the crossing over that took ~~over~~ place in prophase I, they aren't genetically identical to each other.

During Interphase II :-

No DNA replication takes place. The genome was already replicated in Interphase I, which is why the chromosomes entering in meiosis

still have 2 chromatids each.

Q: 3 (iii)

A: 3 (iii)

DNA (in its genes) contains the information needed for the synthesis of proteins. The flow of this information from DNA to RNA and then from RNA to protein, is known as the central dogma of life or molecular biology. Protein synthesis takes place in 2 sequential steps, transcription and translation, and together these 2 are called gene expression.

Transcription :-

is the first step. Both DNA strands of a gene unwind from each other and the nucleotide sequence of only 1 strand, the coding strand, is copied in the form of an mRNA molecule.

Once the gene has been copied,
2 DNA strands coil together
again into their normal double-
stranded state.

Translation ::

is the second step.
The mRNA is moved into the
cytoplasm and binds to a ribosome.
The ribosome reads the code
carried on the mRNA and links
amino acids together in the
sequence those codes specify,
building a polypeptide chain.

Where each take place ::

transcription :- in the nucleus

translation :- in cytoplasm, at
ribosome

mRNA :- carries message of DNA
from nucleus out to the
cytoplasm from protein synthesis.

tRNA :-

transfers the amino acids
to the ribosome as per the
sequence of codes in the mRNA.

Q: 3(iv)(a)
A: 3(i)(or)

Specificity :-

An enzyme is specific because it catalyses only one type of reaction and acts only on one particular substrate. The reason is active site. It is a cavity of a specific shape or charge and only a substrate whose shape and charge fit it can bind. Example: - Amylase acts on starch only. It will not act on proteins or fats.

Optimum pH :- Pepsin has an optimum pH of 2.00 works best at stomach at acidic medium while trypsin optimum pH is 8.00 works best at small intestine at alkaline medium.

© Temperature :-

far above the optimum, the atoms of enzyme vibrate vigorously that the bonds holding its globular structure breaks. The globular structure and with it, the active site is changed. It is permanent and it is denaturation.

far below the optimum, the enzyme is merely inactive because there is not enough energy of activation for the reaction to proceed. It's structure hasn't been damaged so it regains its activity as soon as the normal temperature is restored.

MCQs

- | | |
|------|-------|
| 1. C | 10. C |
| 2. D | 11. B |
| 3. B | 12. A |
| 4. C | |
| 5. D | |
| 6. B | |
| 7. C | |
| 8. B | |
| 9. C | |